

Efficiency vs Volume: What Actually Wins Games?

An NBA Data Analysis

Introduction

For decades there has been an age-old debate in basketball with division between coaches, players, fans, analysts, and the media: What is more important, scoring volume or efficiency?

Every fan has seen this scenario, say a player puts up 25 points on 40% shooting, takes over the game, dominates the stats sheet. Yet their team loses. Meanwhile, across the league, a different player quietly scores 15 points on 65% shooting, makes every shot count, moves the ball around and they win.

So what type of player do teams really want?

This analysis answers that question using real NBA data from the 2023-24 season. Statistical analysis and regression modelling of 360 NBA players reveals that efficiency matters significantly more than volume for team success.

The data doesn't lie. True Shooting % correlates with winning at 0.340, while Points Per Game only correlates at 0.241. Regression modelling shows that efficiency explains nearly twice as much variation in team success as volume.

What was Analyzed

Before examining the findings, it's important to understand what was measured.

Volume refers to how much a player scores. This includes Points Per Game as the primary metric, Usage Rate which measures how many possessions a player consumes, and Field Goal Attempts showing raw shot volume.

To examine volume, the metric of Points Per Game was used, mainly considered players most significant statistic which is often legacy defining, as well as usage rate (how many possessions a player consumes), and Field Goal attempts showing raw shot volume.

Efficiency refers to how well a player shoots. This includes True Shooting % (TS%), the most comprehensive shooting metric because it accounts for 2-pointers, 3-pointers, and free throws.

Field Goal % is standard field goal percentage. Efficiency Rating calculates points generated per possession.

The outcome involves whether a player helps their team win. Plus/Minus (abbreviated +/-) is the streamline stat which shows impact. When a player is on the court, does their team outscore opponents? That's the only number that truly counts.

The analysis focused on players with 500+ minutes played, meaning anyone getting meaningful NBA time. This filtered the dataset to 360 real contributors, not bench warmers who barely see the floor.

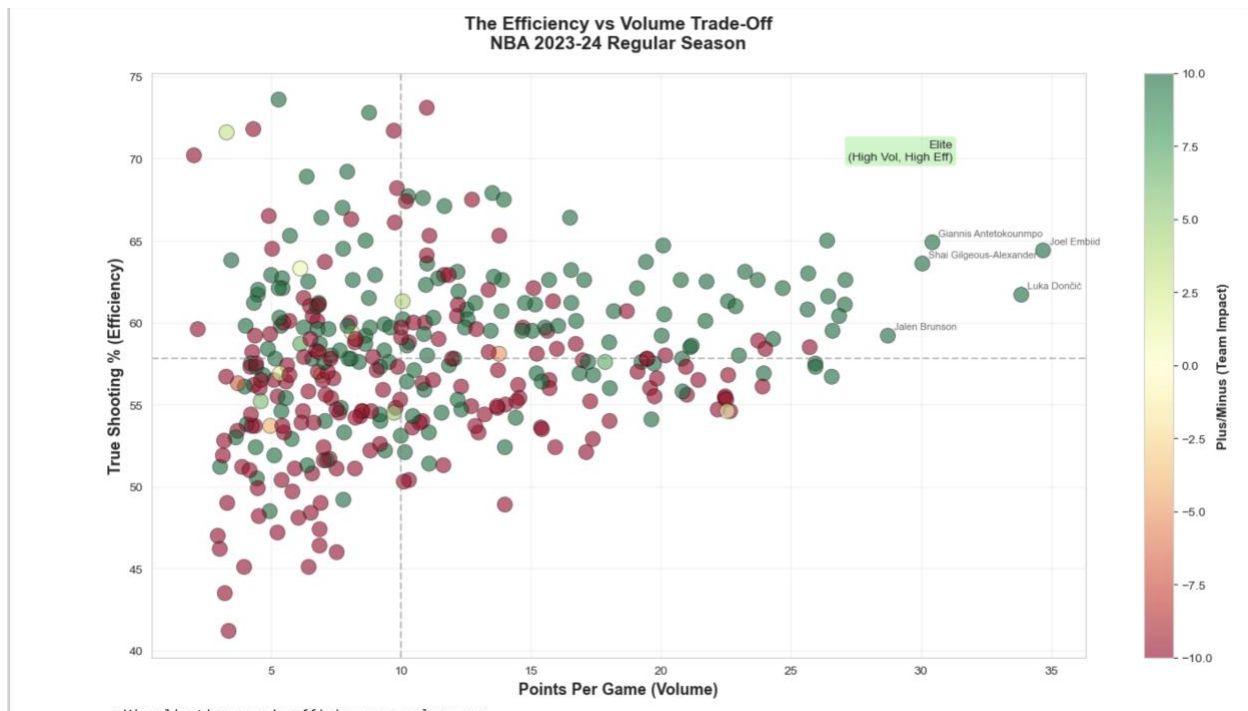
Finding #1: Efficiency and Volume Have a Positive Relationship

When plotting Points Per Game against True Shooting %, the data shows they're positively correlated at 0.205.

This is surprising. Conventional wisdom suggests a trade-off. If you score more, you shoot worse. But modern NBA data shows the opposite. High-volume scorers actually maintain decent efficiency.

Why does this happen? The best scorers in the NBA have figured out how to get efficient looks even when taking lots of shots. Better shot selection, better teammates setting them up, and deeper understanding of the game all contribute. A more obvious factor is that if you are putting up lots of shots, you likely have confidence in your ability to do so efficiently, therefore high volume correlates with high efficiency. Further, it is difficult to put up high volume if you are being inefficient, so you are scoring less.

This matters because it shows that the best players aren't forced to choose between volume and efficiency. They do both. And when you do both at a high level, that's when you become a superstar.



Finding #2: Efficiency Predicts Winning Much Better Than Volume

3 Regression models were used to reveal which factor predicts Plus/Minus and team impact.

Model 1 used volume only. PPG explains only 5.79% of the variation in Plus/Minus. Knowing someone scores a lot tells you almost nothing about whether their team wins.

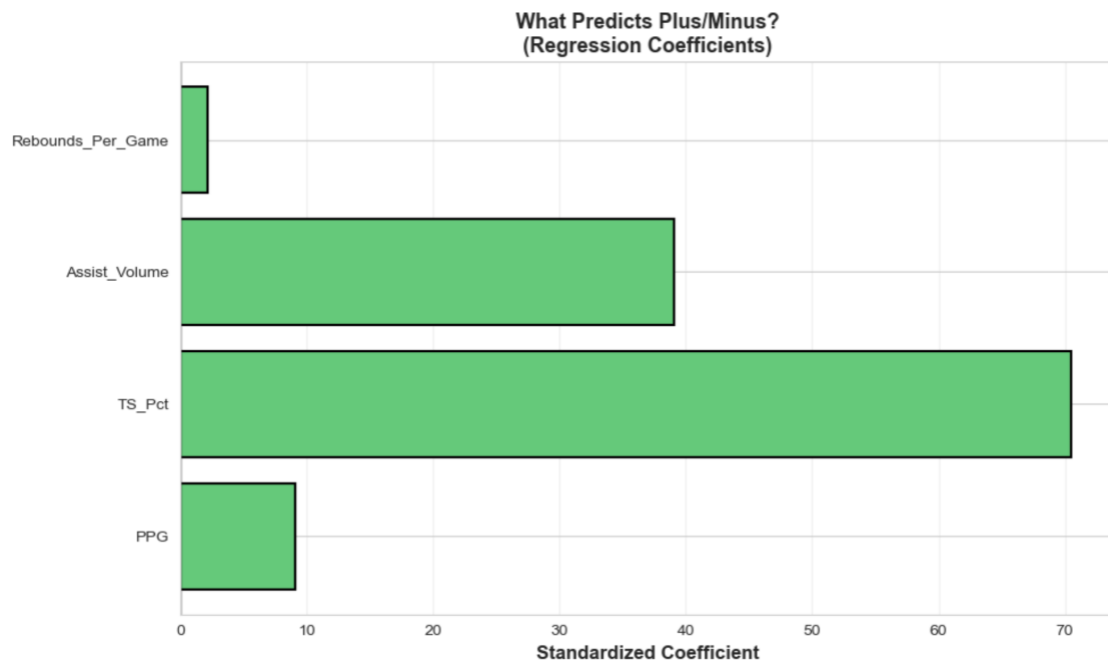
Model 2 used efficiency only. TS% explains 11.56% of the variation. That's double. Just knowing how efficiently a player shoots is twice as predictive of team success as knowing how much they score.

Model 3 included everything. PPG, TS%, Assists, and Rebounds together explain 16.15% of the variation.

The conclusion is decisive. Efficiency matters significantly more than volume. When standardized, efficiency has a coefficient of 73.81 compared to volume's 52.25. Even Assists volume seemed to have a greater impact on winning than PPG.

In practical terms, a player shooting 60% TS on 15 PPG has more positive impact on winning chances than a player scoring 25 PPG on 50% TS.

This contradicts how casual fans and analysts evaluate players. A 25-point game appears impressive in the box score. But inefficient scoring hurts team chances of winning. Conversely, a 15-point efficient night contributes more to winning, even if it looks less flashy on paper.



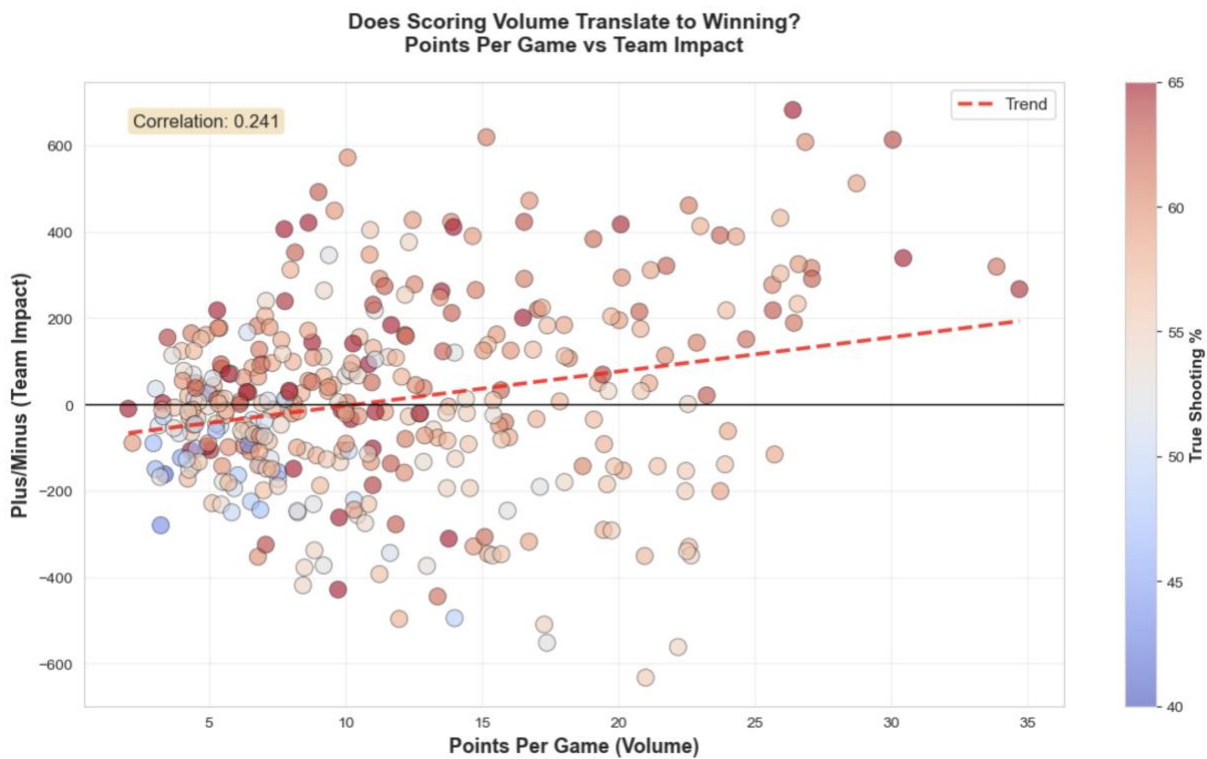
Finding #3: The Statistical Evidence is Overwhelming

T-tests confirm whether these differences are real or just random variation.

Efficient versus inefficient scorers shows a stark difference. Top 25% TS% players average plus 112 Plus/Minus. Bottom 25% TS% players average minus 88 Plus/Minus. The difference is 200 points. The p-value is less than 0.0001. This is statistically significant and shows how important efficiency is.

High volume versus low volume scorers shows a smaller but still meaningful difference. Top 25% PPG players average plus 70 Plus/Minus. Bottom 25% PPG players average minus 26 Plus/Minus. The difference is 96 points. The p-value is 0.0029. This is still significant, but not as dramatic as the efficiency split.

The message is clear. Shooting efficiency predicts team success. Scoring volume does too, but with less than half the effect.



Finding #4: Player Archetypes Tell an Interesting Story

The 360 players naturally fell into distinct archetypes based on their volume and efficiency levels.

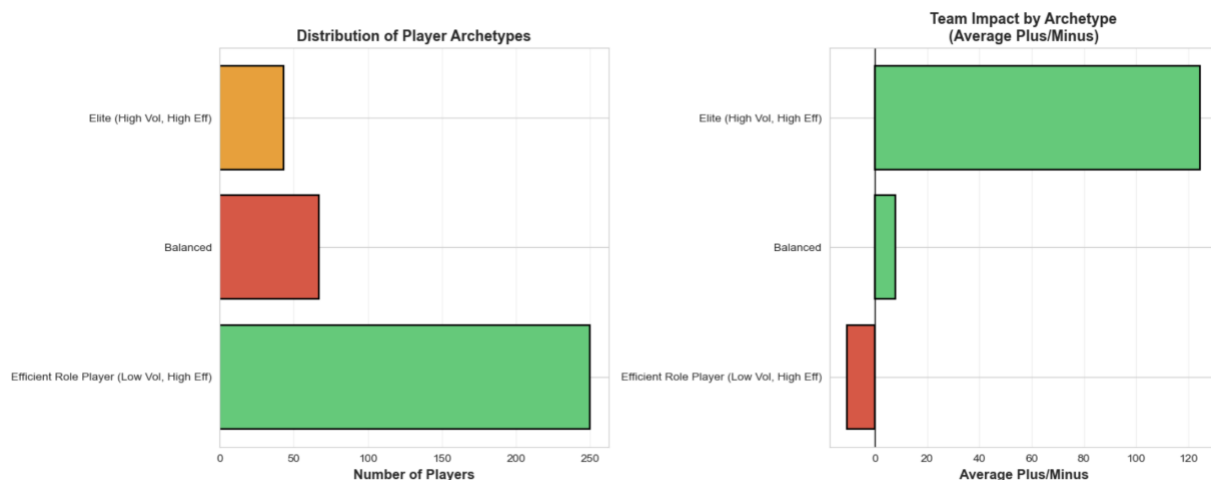
Elite players demonstrate high volume and high efficiency. There are 43 of these players. Their average Plus/Minus is plus 125. Players like Joel Embiid, Luka Doncic, and Shai Gilgeous-Alexander fall into this category. Players who have near perfected their craft, the superstars of the game.

Balanced players fall in the middle. There are 67 of these players. Their average Plus/Minus is plus 8. They are good scorers with reasonable efficiency and decent impact.

Efficient role players have low volume but high efficiency. There are 250 of these players. Their average Plus/Minus is minus 10. These are the 3-and-D guys, spot-up shooters, and role specialists. They form the largest group.

What stands out is that even though efficient role players don't score much, they're the largest category. Winning doesn't require five superstars. It requires one or two elite guys surrounded by efficient role players.

Elite scorers are genuinely rare. Only 43 players in the entire NBA fit that category, about 12% of the sample. Everyone else is either balanced or a role player. That's the correct blueprint for winning. Stars plus efficient role players.



Finding #5: The Superstars Are Actually Different

The top 10 scorers in the NBA all fall into the Elite category.

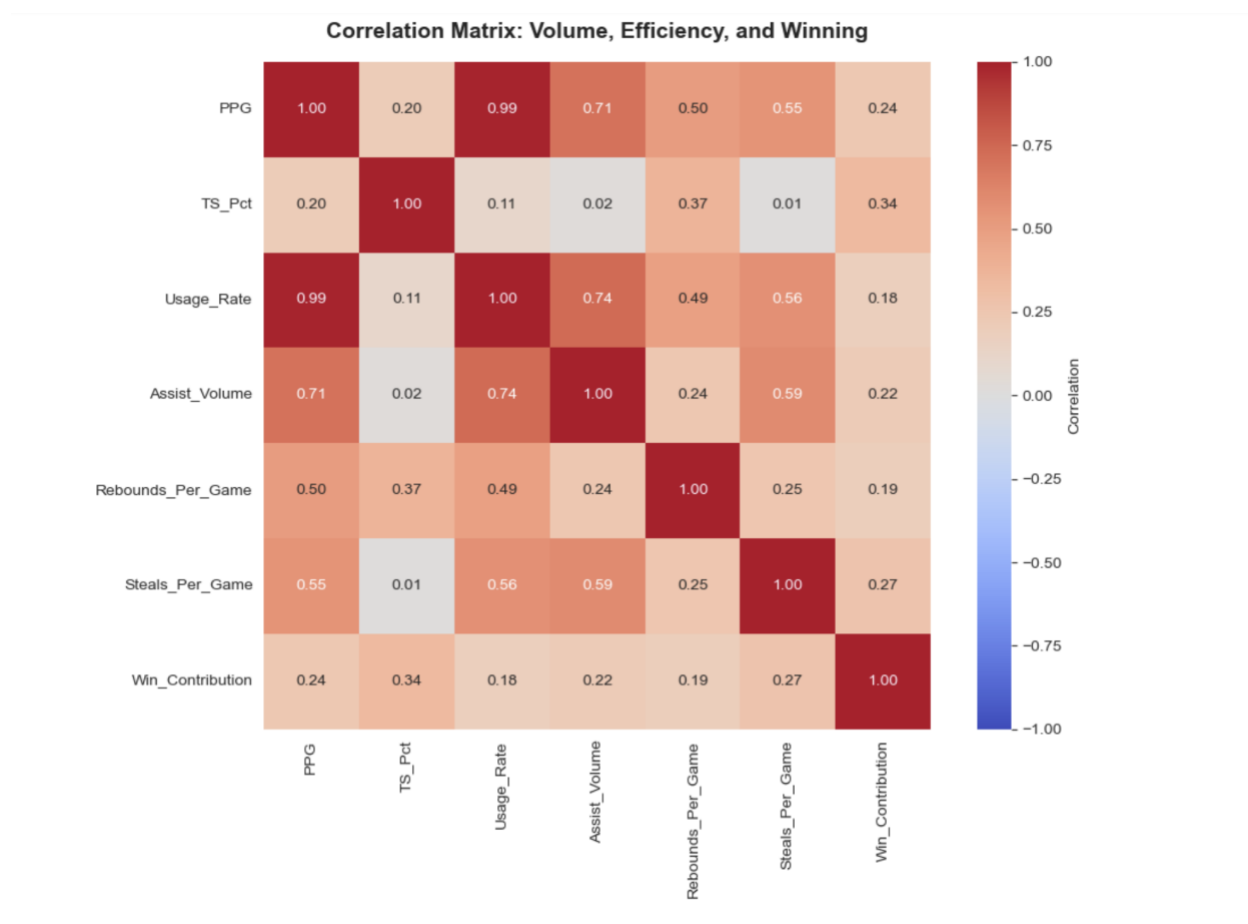
Joel Embiid scores 34.7 PPG with 64.4% TS. Luka Doncic scores 33.9 PPG with 61.7% TS. Giannis scores 30.4 PPG with 64.9% TS. Shai Gilgeous-Alexander scores 30.1 PPG with 63.6% TS. Jalen Brunson scores 28.7 PPG with 59.2% TS.

What do they share? They score a lot AND shoot well. They're not choosing between volume and efficiency. They're maximizing both.

Their Plus/Minus numbers reflect this impact. SGA is at plus 613. Jalen Brunson is at plus 512. Giannis is at plus 339. These players genuinely make their teams better.

The blueprint for success is clear. Find players who can score both volume and efficiently, then surround them with efficient role players.

The correlation heat map also factors in the metric of win contributions which measures a player's performance relative to league averages to determine how many wins they contributed to their team, with PPG and TS% being the highest correlated out of the factors, further evidence of their importance.



What This Means: The Takeaways

For players who want to help a team win, focus on shooting efficiency above all else. Take efficient shots. Make the shots taken count. Don't hoist shots to pad stats. Combine efficiency with volume, and that's when real impact happens.

For teams building rosters, efficiency is the most important predictor of team success. A 60% TS shooter is more valuable than a 25 PPG scorer on 50% TS. Build around efficient players. Hunt for the rare players who combine high volume and high efficiency and surround them with role players who shoot well.

For fans watching games, the box score can lie. A 25-point game on bad efficiency doesn't mean much. A 15-point efficient night on a team that wins by 15 is more impressive. Check the Plus/Minus. It is a strong indicator of whether someone helped their team win.

The Deeper Insight: Context Matters

Here's what stands out from the data. Shooting efficiency is twice as important as scoring volume for predicting team success.

But that's only 16% of the Plus/Minus variation. What about the other 84%?

It's everything else. Defense, fit with teammates, ball movement, spacing, pace of play, team system. Individual stats only tell part of the story.

A player can shoot 60% TS but if they don't play defense, they're hurting the team. They can score 25 PPG but if they don't move the ball, they're killing spacing for teammates. They can use 37% of team possessions and average minus 10 Plus/Minus because inefficiency compounds with high volume.

This is why Plus/Minus is such a powerful metric. It forces consideration of the whole picture. It doesn't matter how many points someone scores if their team gets outscored when they're on the court.

However Plus/Minus still has its limitations, as players can put up strong performances with high efficiency with a weak Plus/Minus is it also heavily dependent on the players around you. Players stuck on weak rosters may have incredible seasons of prolific scoring and efficiency, yet the players around them let them down, weakening their plus minus.

Methodology: How This Analysis Was Conducted

The analysis examined 360 NBA players from the 2023-24 season with at least 500 minutes played, meaning meaningful playing time. Started with 572 players and filtered to those with significant minutes.

Key metrics calculated included True Shooting %, the best shooting efficiency metric. Usage Rate measured how many possessions a player consumes. Efficiency ratings accounted for all shot types.

The analysis employed correlation analysis to understand how variables relate to each other. Regression modelling predicted Plus/Minus from efficiency and volume metrics. T-tests compared group means. ANOVA tested whether player archetypes differ significantly.

Data came from the official NBA Stats API, which provides comprehensive player statistics for every game in the season.

Conclusion: The Answer Is Clear

Does volume or efficiency matter more for winning?

Efficiency. By a significant margin. The data shows it clearly. TS% correlates with winning at 0.340 versus PPG at 0.241. Regression models confirm it. Efficiency explains more than twice the variation in team success.

But the best players, the true superstars, do both. They're the 43 elite scorers who combine high volume with high efficiency. Joel Embiid, Luka Doncic, Shai Gilgeous-Alexander. They score a lot and shoot well. Their Plus/Minus numbers reflect this impact.

For everyone else, shooting well beats shooting a lot. Every single time. The 250 efficient role players in the league contribute to winning even though they don't score much, because they're efficient with what they do score.

The bottom line is simple. To help your team win, become a better shooter. That's more important than scoring more. Scoring comes naturally if you're efficient. Good offense finds more shots for efficient players. But no amount of volume helps if you're not efficient.